

```
%nprocshared=64
%mem=320GB
%chk=DPRhaC8C8.chk
# opt=loose freq apfd/6-311+g(2d,p) scrf=(solvent=water) scf=xqc
```

DPRhaC8C8

-1 1

C	-1.54737729	0.34034011	-4.46868345
C	-0.70010002	-0.20135045	-3.26291731
C	0.43310688	0.69030601	-3.04861301
O	-0.23625926	2.01829043	-2.62176342
C	-1.63675057	2.40172334	-3.19677845
C	-1.59876146	1.90294990	-4.52417211
C	-1.75750768	3.93438550	-3.22638215
O	-2.67381755	2.29099611	-5.37824197
O	1.38656142	0.17468750	-2.00249430
O	-1.03701922	0.00298948	-5.75395713
O	-0.39083435	-1.57267304	-3.52302832
C	0.92423340	-0.26243115	-0.62836422
C	2.13862384	-0.85643753	0.12298754
C	0.51110892	0.89026845	0.38923609
C	2.89791627	-2.01250541	-0.45816554
O	4.14007627	-2.58168944	0.14776612
O	2.52295694	-2.46669165	-1.54410644
C	5.05424433	-3.70408480	-0.42113468
C	-0.68533152	1.79406033	0.07565585
C	-0.83346887	2.80493435	1.24085239
C	-2.02183755	3.75186892	0.99533667
C	6.13342236	-3.97877110	0.66086694
C	6.86144981	-2.74062338	1.09175516
O	7.91136519	-2.84867797	2.00460021
O	6.58328326	-1.64387868	0.59732749
C	4.45845514	-5.16226829	-0.68028401
C	3.34499210	-5.15776399	-1.73806628
C	2.73331171	-6.52373961	-2.03166997
C	1.65818325	-6.26350499	-3.10679190
H	0.14741950	-1.03268393	-0.71314560
H	5.57166589	-3.36439849	-1.33045427
H	-2.56458529	-0.04185239	-4.38313950
H	-1.32671786	-0.16416284	-2.37181355
H	0.96600175	0.83567325	-3.98828835
H	-2.44360726	1.94768065	-2.62151550
H	-0.67156100	2.24860639	-4.98125133
H	-2.73187806	4.21493004	-3.62625584
H	-0.97265750	4.34904120	-3.85902301
H	-1.65370725	4.32713852	-2.21487953
H	-3.58254636	1.95730247	-4.97152257
H	-0.06996011	0.39763678	-5.86170925
H	-1.28132746	-2.12120910	-3.61624900

H	2.85464325	-0.04472238	0.25157155
H	1.77554305	-1.17931460	1.09870402
H	0.29932565	0.40718788	1.34311252
H	1.37842538	1.54178032	0.49597999
H	-0.51168023	2.32808694	-0.85856089
H	-1.59077996	1.19349424	-0.01142755
H	-0.99828366	2.25819473	2.16929817
H	0.08081645	3.39278671	1.32216528
H	-1.85956817	4.30234508	0.06865276
H	-2.93866867	3.16775205	0.91578497
H	5.64712896	-4.41645503	1.53267624
H	6.85872851	-4.68463253	0.25615846
H	5.26294467	-5.81392281	-1.02122592
H	4.05134614	-5.54574789	0.25529284
H	2.54985817	-4.49852248	-1.38983869
H	3.76145749	-4.76669924	-2.66635757
H	3.49240598	-7.20915587	-2.40860569
H	2.28080613	-6.94097973	-1.13208732
H	0.92377902	-5.56277752	-2.70963963
H	2.13537818	-5.83095328	-3.98615773
C	0.93843041	-7.56161807	-3.51722861
H	0.39301764	-7.94670544	-2.68108711
H	0.26182818	-7.35572143	-4.32017070
H	1.66035766	-8.28436268	-3.83560148
C	-2.15027784	4.74656718	2.16395913
H	-2.98597069	5.39239206	1.99241413
H	-2.29743220	4.20762429	3.07653031
H	-1.25667153	5.33080644	2.23489756